C++ Programming

Trainer: Nilesh Ghule

What is class? → user defined data type (like struct inc)

- Building block that binds together data & code → fns
- Program is divided into different classes
- Class has
    - Variables (data members)
    - Functions (member functions or methods)
- By default class members are private(not accessible outside class scope)
- Classes are stand-alone components & can be distributed in form of libraries
- Class is blue-print of an object

What is object?

fn → msgs
behavior

t5.display(); → 10:10:10
t6.display(); → 20:20:20

- Object is an instance of class
- Entity that has physical existence, can store data, send and receive message to communicate with other objects

time t5(10,10,10);
time t6(20,20,20);
time t7(20,20,20);

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- Object has
  - Data members (***state*** of object)
  - Member function (***behavior*** of object)
  - Unique address(***identity*** of object)
- The values stored in data members of the object called as 'state' of object
- Behavior is how object acts & reacts, when its state is changed & operations are done

fns → messages or operations or methods

- Behavior is decided by the member functions. Objects behaves differently depending upon the state of the object.
- Operations performed are also known as messages

Procedure oriented vs Object oriented

- Emphasis on steps or algorithm
- Programs are divided into small code units i.e. functions
- Most functions share global data & can modify it
- Data move from function to function
- Top-down approach

- Emphasis on data of the program
- Programs are divide into small data units i.e. classes
- Data is hidden & not accessible outside class
- Objs communicate with each other
- Bottom- up approach

cin and cout

- C++ provides an easier way for input and output.
- The output:
  - cout << “Hello C++”;
- The input:
  - cin >> var;
- Both cin and cout are buffered IO.

```
#include <iostream>
using namespace std;
int main() {
        int a, b, c;
        cout << "enter two numbers: ";
        cin >> a >> b;
        c = a + b;
        cout << "result: " << c << endl;
        return 0;
}
```

Inline functions

- C++ provides a keyword *inline* that makes the function as inline function.
- Inline functions get replaced by compiler at its call statement. It ensures faster execution of function just like macros.
- Advantage of inline functions over macros: inline functions are type-safe.
- Inline is a request made to compiler.

Variable declaration & Data Types

- In C, variable should be declared at the start of the block.
- This restriction is removed in C++. We can declare the variables anywhere in function.
- In C, `const int x;` is valid statement.
- In C++, it is not valid. Constant variables (local or global) should be initialized at the point of declaration.
  `const int x=10;`

- C++ supports all data types provided by C language. i.e. int, float, char, double, long int, unsigned int, etc.
- C++ add two more data types:

1. ***bool*** :- it can take *true* or *false* value. It takes one byte in memory.
2. ***wchar_t*** :- it can store wide characters (i.e. Unicode). It takes 2 or 4 bytes in memory - depends of C++ implementation.

Thank You!

Nilesh Ghule

<nilesh@sunbeaminfo.com>